Jonathan Chan

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EDUCATION

University of Oxford, Merton College — **Oxford, United Kingdom**
DPhil in Medical Sciences (Bioinformatics/Cardiovascular Genomics) — 2022 – Present (Oct 2026)

University of Oxford, St Hilda's College — **Oxford, United Kingdom**
MBiochem in Molecular and Cellular Biochemistry – 1st Class (Rank: 2nd of 97) — 2018 – 2022

HONOURS

- Radcliffe Department of Medicine & Clarendon Fund DPhil Scholarship (~£175k), 2022
- Gibbs Prize for Proxime Accessit in Biochemistry (2nd in Cohort), 2022
- Department of Biochemistry Project Prize for Master's Research (3rd in Cohort), 2022

RESEARCH EXPERIENCE

Dart Biosciences Ltd, London, United Kingdom
Computational Biologist (Part-Time), Oct 2024 – Present

- Rapidly prototyped MVP for cell-specific marker gene identification (*cellxgene*, *scVI*, *cellPLM*), ultimately driving a strategic "no-go" decision by identifying commercial viability constraints.
- Drove early-stage project derisking by benchmarking SOTA sc-transcriptome → surfaceome (CITE-seq) prediction models leading to data-backed "no-go" recommendation for R&D resource redirection.
- Spearheaded multi-omic integration (public scRNA-seq/ATAC-seq datasets) to identify activation-dependent genes and cis-regulatory elements for downstream *in vitro* validation.
- Leading the *in silico* design of synthetic cis-regulatory elements (CREs) to drive cell-type specific *in vivo* payload expression, for precision gene delivery targeting and minimizing off-target effects.

University of Oxford Radcliffe Department of Medicine, Oxford, United Kingdom
Doctoral Student (Prof. Hugh Watkins & Prof. Anuj Goel), Sep 2022 – Present

- Engineering bioinformatic and ML pipelines integrating genomic, proteomic, clinical, and imaging data across population cohorts (UK Biobank, *All of Us*; N > 1.1M) and a 2,500+ patient registry.
- Building multifactorial predictive models for hypertrophic cardiomyopathy (HCM) to enable improved patient risk stratification, enhanced disease surveillance, and individualized prognosis.
- Identifying novel disease progression biomarkers via population-scale analyses and statistical genetic approaches (colocalisation, Mendelian randomisation).

University of Oxford Ludwig Institute for Cancer Research, Oxford, United Kingdom
Master's Student (Prof. Yang Shi), Sep 2021 – Jun 2022

- Drove pre-clinical target identification by bridging wet-lab execution and computational biology via genome-wide CRISPR-Cas9 knockout screening and resulting NGS data analysis.
- Identified and prioritized novel genetic vulnerabilities in pediatric glioma model, successfully advancing the top identified therapeutic target (ASCL1) to *in vivo* validation via mouse models.

SKILLS

Tech
- R: *tidyverse*, *ggplot2*, *markdown*
- Python: *pandas, NumPy, seaborn*, *scikit-learn, scikit-survival*
- HPC: *SLURM*

- Cloud Computing: *DNAnexus, GCP*

Bioinformatics
- GWAS (*REGENIE, PLINK*, *SNPTEST*)
- Post-GWAS analyses (*LDSC*, *FINEMAP*, *coloc*)
- Polygenic risk score modelling (*PRS-CS*, *SBayesRC*)
- Rare-variant association analysis (*REGENIE*, *SAIGE-GENE+*, *VEP*)
- Mendelian randomisation (*TwoSampleMR, MR-PRESSO*)
- Pipelining (*snakemake*, *Bash*, *Docker*, *WDL*)
- Single-cell transcriptomic/multi-omic analysis (*scanpy*, *scVI*, *cellxgene, DESeq2, muon*)

Population-Scale Data Analysis
- Data analysis in patient & population datasets (*HCMR*, *UK Biobank*, *All of Us*)
- Systematic review (*PRISMA*) and meta-analysis (*meta*)
- Survival/TTE analysis (*Kaplan-Meier, Cox regression, RSF, GBST*)

PUBLICATIONS/PRESENTATIONS

<u>Chan, J.H.</u>, Grace, C., Mahidi, M., Clarke, R., Ho, C., Neubauer, S., Kramer, C., Watkins, H., Goel, A.
Biobank-Scale Plasma Proteomics Identifies Novel Biomarkers in Hypertrophic Cardiomyopathy,
Circulation: Precision & Genomic Medicine *(Accepted)*

Jiao, A.L., Maghrouni, A., Velazquez, A.G., Annett, A., Wen, T., <u>Chan, J</u>. … Shi, Y.
H3K27M co-opts ASCL1 to maintain progenitor states and drive gliomagenesis, Nature Communications *(Under Review)*

Prioritising modifiable risk factors and biomarkers for hypertrophic cardiomyopathy: a Mendelian randomization approach
American Society of Human Genetics Annual Meeting (Boston, US), Oct 2025

Uncovering how Sarcomeric Variants affect Disease Heterogeneity in Hypertrophic Cardiomyopathy: A Cardiac Magnetic Resonance Imaging Study in the HCM Registry
International Society of Heart Research XXV World Congress (Nara, Japan), May 2025

Integrating Rare and Common Genetics for Risk Prediction in Hypertrophic Cardiomyopathy
Genomics England Research Summit (London, UK), Jul 2024

EXTRACURRICULAR

Peer Review Experience
- Nature Cardiovascular Research
    - NCVR-2025-03-0348 (Statistical genetics, Mendelian randomisation), Apr 2025
- Circulation: Heart Failure
    - CIRCHF/2026/014278 (Hypertrophic cardiomyopathy, plasma proteomics), Feb 2026
- Nature Communications
    - NCOMMS-26-005224 (Cardiovascular genomics, statistical genetics), Mar 2026

Oxford University Biotech Society, Oxford, United Kingdom
President, Jun 2023 – Jun 2024
- Managed team of 11 across all society facets e.g., operations, finance, communications etc.
- Spearheaded inaugural OxBioHack hackathon in collaboration with Oxford CompSoc and OPMS.